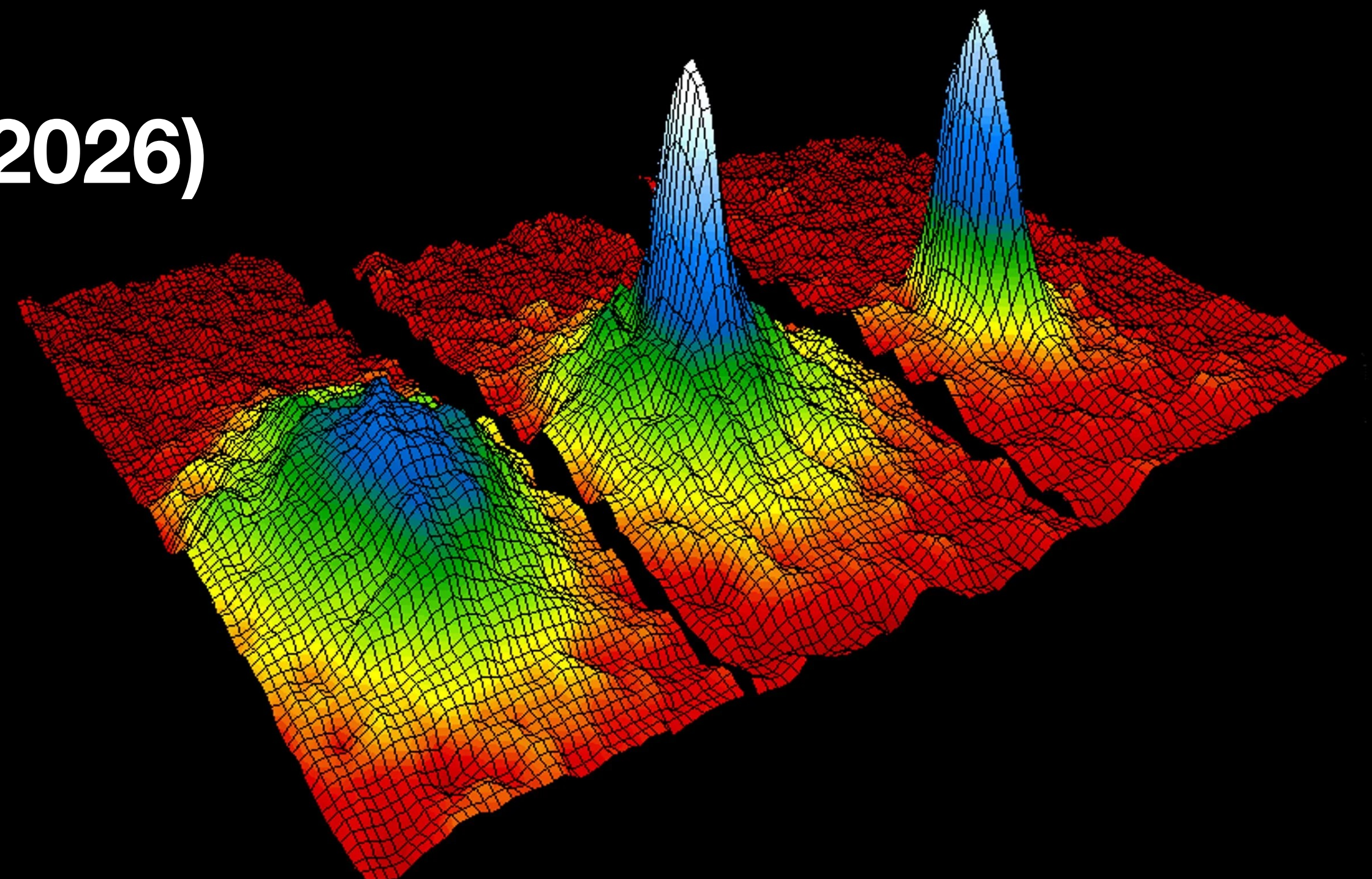


Introduction to Quantum Many-Body Physics with Bose-Einstein Condensate

UNIST Tutorial Session II (May 7th 2026)

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States in Second Quantization (Recap)

First Quantization

Single particle:

$$|\psi\rangle = \sum_i c_i |u_i\rangle \in \mathcal{H}$$

N-**identical** particles:

$$|\Psi\rangle \in \mathcal{H}^{\otimes N}$$

N-**indistinguishable**
particles:

$$|\Psi\rangle \in \mathcal{S}_{\pm} \mathcal{H}^{\otimes N}$$

Second Quantization

N-**indistinguishable** particles:

$$|\Psi\rangle = \sum_{\{n_i\}} c_{\{n_i\}} |\{n_i\}\rangle \in \mathcal{F}$$

Fock space: $\mathcal{F} = \bigoplus_{N=0}^{\infty} \mathcal{S}_{\pm} \mathcal{H}^{\otimes N}$

Bosonic (fermionic) **creation and annihilation operators:** $\mathcal{F}_N \rightarrow \mathcal{F}_{N\pm 1}$

$$[\hat{b}_i, \hat{b}_j^{\dagger}] = \delta_{ij} \quad \{\hat{c}_i, \hat{c}_j^{\dagger}\} = \delta_{ij}$$

One-body Operator (Recap)

$$\hat{F} = \sum_{q=1}^N \hat{f}_q \quad \hat{F} : \mathcal{H}^{\otimes N} \rightarrow \mathcal{H}^{\otimes N}$$

Second
Quantization

$$\hat{F} = \sum_{k,l} f_{kl} \hat{b}_k^\dagger \hat{b}_l \quad \hat{F} : \mathcal{F}_N \rightarrow \mathcal{F}_N$$

$$f_{kl}^{(q)} = \langle u_k^{(q)} | \hat{f}_q | u_l^{(q)} \rangle = f_{kl}$$

Examples:

- Particle number operator

$$\hat{N} = \sum_i \hat{b}_i^\dagger \hat{b}_i$$

- Non-interacting Hamiltonian

$$\hat{H} = \sum_i \epsilon_i \hat{b}_i^\dagger \hat{b}_i$$

Two-body Operator

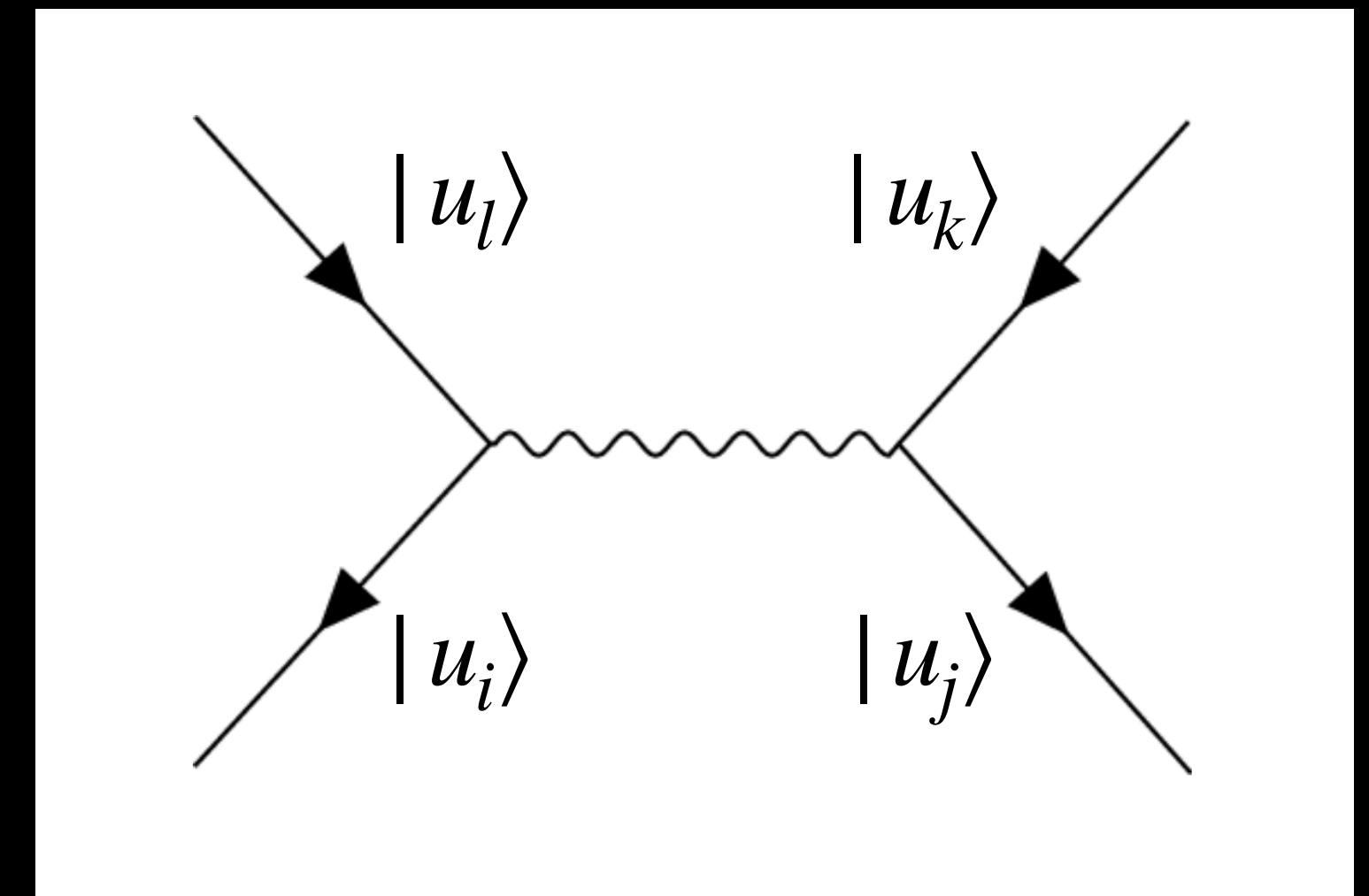
$$\hat{F} = \frac{1}{2} \sum_{\substack{p, q = 1 \\ p \neq q}}^N \hat{f}_{pq} \quad \xrightarrow[\text{Second Quantization}]{\mathcal{F}_N \rightarrow \mathcal{F}_N}$$

$$\hat{F} = \frac{1}{2} \sum_{ijkl} f_{ijkl} \hat{b}_i^\dagger \hat{b}_j^\dagger \hat{b}_l \hat{b}_k$$

$$\hat{f}_{pq} = \sum_{ijkl} f_{ijkl} \left(|u_i\rangle_p \otimes |u_j\rangle_q \right) \left({}_p\langle u_k| \otimes {}_q\langle u_l| \right)$$

Order matter for fermions

- The formalism can be generalized to any N -body operator
- But in physics we typically only deal with **pair-wise interactions**



Hamiltonian of Pairwise Interacting Particles

$$\hat{H} = \hat{T} + \hat{V} + \hat{W}_{\text{interaction}} \quad \hat{T} = \sum_{q=1}^N \hat{t}_q \quad \hat{V} = \sum_{q=1}^N \hat{v}_q \quad \hat{W} = \sum_{\substack{p,q=1 \\ p \neq q}}^N \hat{w}_{pq}$$

(Ignore spin)

In second-quantized form with a generic single-particle basis $\{ |u_i\rangle \} \in \mathcal{H}$:

$$\hat{H} = \sum_{ij} (t_{ij} + v_{ij}) \hat{b}_i^\dagger \hat{b}_j + \frac{1}{2} \sum_{ijkl} w_{ijkl} \hat{b}_i^\dagger \hat{b}_j^\dagger \hat{b}_l \hat{b}_k$$

If one instead chooses single-particle energy eigenbasis $\{ |E\rangle_i \} \in \mathcal{H}$:

$$\hat{H} = \sum_i \varepsilon_i \hat{b}_i^\dagger \hat{b}_i + \frac{1}{2} \sum_{ijkl} w_{ijkl} \hat{b}_i^\dagger \hat{b}_j^\dagger \hat{b}_l \hat{b}_k$$

The Field Operator $\hat{\Psi}(\mathbf{r})$

Recall the change of basis in second quantization: $\{ |u\rangle_i \} \rightarrow \{ |v\rangle_i \} \in \mathcal{H}$

$$\hat{b}_{v_j} = \sum_i \langle v_j | u_i \rangle \hat{b}_{u_i}$$

Consider position eigenbasis $\{ |\mathbf{r}\rangle \}$:
$$\hat{b}_{\mathbf{r}} = \sum_i \underbrace{\langle \mathbf{r} | u_i \rangle}_{u_i(\mathbf{r})} \hat{b}_{u_i}$$

Change into standard notation: $\hat{b}_{\mathbf{r}} \rightarrow \hat{\Psi}(\mathbf{r})$ **(WARNING ⚠: NOT QUANTUM STATE)**

$$\hat{\Psi}(\mathbf{r}) = \sum_i u_i(\mathbf{r}) \hat{b}_i$$

$$\hat{\Psi}^\dagger(\mathbf{r}) = \sum_i u_i^*(\mathbf{r}) \hat{b}_i^\dagger$$

Field operator $\hat{\Psi}(\mathbf{r}) \rightarrow$
Example **quantum field**, i.e.
operator-valued field.

The Field Operator $\hat{\Psi}(\mathbf{r})$

By orthogonality $\int u_i^*(\mathbf{r})u_j(\mathbf{r}) d\mathbf{r} = \delta_{ij}$ we can invert the relation:

$$\hat{b}_i = \int u_i(r)\hat{\Psi}(\mathbf{r}) d\mathbf{r}$$

$$\hat{b}_i^\dagger = \int u_i^*(r)\hat{\Psi}^\dagger(\mathbf{r}) d\mathbf{r}$$

Commutation relation (boson):

$$[\hat{\Psi}(\mathbf{r}), \hat{\Psi}^\dagger(\mathbf{r}')] = \delta(\mathbf{r} - \mathbf{r}')$$

$$[\hat{\Psi}(\mathbf{r}), \hat{\Psi}^\dagger(\mathbf{r})] = 0$$

Relation with operator for particle density: $\hat{n}(\mathbf{r}) = \hat{\Psi}^\dagger(\mathbf{r})\hat{\Psi}(\mathbf{r}) \rightarrow$ an observable
(WARNING ⚠: NOT “DENSITY OPERATOR” IN QUANTUM STATE SENSE)

$$\int \hat{\Psi}^\dagger(\mathbf{r})\hat{\Psi}(\mathbf{r}) d\mathbf{r} = \sum_{ij} \underbrace{\left(\int u_i^*(\mathbf{r})u_j(\mathbf{r}) d\mathbf{r} \right)}_{\delta_{ij}} \hat{b}_i^\dagger \hat{b}_j = \sum_i \hat{b}_i^\dagger \hat{b}_i = \hat{N}$$

The Field Operator $\hat{\Psi}(\mathbf{r})$

Consider correlation functions of field operators:

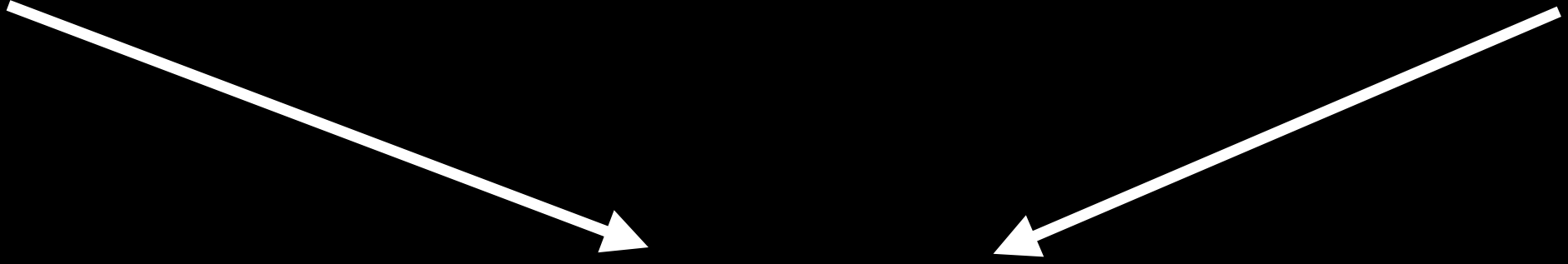
$$G_N(\mathbf{r}_1, \dots, \mathbf{r}_N; \mathbf{r}'_1, \dots, \mathbf{r}'_N) = \langle \hat{\Psi}^\dagger(\mathbf{r}_1) \dots \hat{\Psi}^\dagger(\mathbf{r}_N) \hat{\Psi}(\mathbf{r}'_1) \dots \hat{\Psi}(\mathbf{r}'_N) \rangle$$

where $\langle X \rangle = \text{Tr}(\hat{\rho}X)$ and $\hat{\rho}$ is the **DENSITY MATRIX** (now, means the state).

ALL correlation functions $\{G_N : \forall N\}$ = Information about many-body state $\hat{\rho}$

$$\hat{b}_i^\dagger = \int u_i^*(r) \hat{\Psi}^\dagger(\mathbf{r}) d\mathbf{r}$$

$$\hat{b}_i = \int u_i(r) \hat{\Psi}(\mathbf{r}) d\mathbf{r}$$


$$\langle \hat{F} \rangle = \frac{1}{2} \sum_{ijkl} f_{ijkl} \langle \hat{b}_i^\dagger \hat{b}_j^\dagger \hat{b}_l \hat{b}_k \rangle$$

Pairwise Interacting Hamiltonian in terms of the Field Operator

$$\hat{H} = \sum_{ij} (t_{ij} + v_{ij}) \hat{b}_i^\dagger \hat{b}_j + \frac{1}{2} \sum_{ijkl} w_{ijkl} \hat{b}_i^\dagger \hat{b}_j^\dagger \hat{b}_l \hat{b}_k$$

Consider first kinetic term:

$$\begin{aligned} \sum_{ij} \langle u_i | \hat{t} | u_j \rangle \hat{b}_i^\dagger \hat{b}_j &= \sum_{i,j} \iint d\mathbf{r} d\mathbf{r}' \langle u_i | \mathbf{r} \rangle \langle \mathbf{r} | \hat{t} | \mathbf{r}' \rangle \langle \mathbf{r}' | u_j \rangle \hat{b}_i^\dagger \hat{b}_j \\ &= \iint d\mathbf{r} d\mathbf{r}' \underbrace{\left(\sum_i u_i^*(\mathbf{r}) \hat{b}_i^\dagger \right)}_{\hat{\Psi}^\dagger(\mathbf{r})} \underbrace{\langle \mathbf{r} | \hat{t} | \mathbf{r}' \rangle}_{-\frac{\hbar^2}{2m} \delta(\mathbf{r}-\mathbf{r}') \nabla_{\mathbf{r}}} \underbrace{\left(\sum_j u_j(\mathbf{r}') \hat{b}_j \right)}_{\hat{\Psi}(\mathbf{r}')} \\ &= \int d\mathbf{r} \hat{\Psi}^\dagger(\mathbf{r}) \left(-\frac{\hbar^2}{2m} \nabla^2 \right) \hat{\Psi}(\mathbf{r}) \end{aligned}$$

Similarly for potential term:

$$\sum_{ij} v_{ij} \hat{b}_i^\dagger \hat{b}_j = \int d\mathbf{r} \hat{\Psi}^\dagger(\mathbf{r}) V(\mathbf{r}) \hat{\Psi}(\mathbf{r})$$

Pairwise Interacting Hamiltonian in terms of the Field Operator

Doing the same thing for the pairwise interaction term gives:

$$\hat{H} = \int d\mathbf{r} \hat{\Psi}(\mathbf{r}) \left[-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{r}) \right] \hat{\Psi}(\mathbf{r}) + \iint d\mathbf{r} d\mathbf{r}' U(\mathbf{r}, \mathbf{r}') \hat{\Psi}^\dagger(\mathbf{r}) \hat{\Psi}^\dagger(\mathbf{r}') \hat{\Psi}(\mathbf{r}') \hat{\Psi}(\mathbf{r})$$

Observation:

(1) **U(1) Symmetry:** $\hat{\Psi}(\mathbf{r}) \rightarrow e^{i\phi} \hat{\Psi}(\mathbf{r})$ leaves $\hat{H} \rightarrow \hat{H}$

(2) **Particle number is conserved:** $[\hat{H}, \hat{N}] = \int d\mathbf{r} [\hat{H}, \hat{\Psi}^\dagger(\mathbf{r}) \hat{\Psi}(\mathbf{r})] = 0$

Indeed, the two statements are connected via Noether's theorem:

Symmetry $\leftrightarrow$ Conserved Quantity

Field Operator Evolution

Heisenberg equation of motion:

$$i\hbar\partial_t\hat{\Psi}(\mathbf{r},t) = [\hat{\Psi}(\mathbf{r},t),\hat{H}]$$

$$i\hbar\partial_t\hat{\Psi}(\mathbf{r},t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}) \right] \hat{\Psi}(\mathbf{r},t) + \int d\mathbf{r}' U(\mathbf{r},\mathbf{r}')\hat{\Psi}^\dagger(\mathbf{r}',t)\hat{\Psi}(\mathbf{r}',t)\hat{\Psi}(\mathbf{r},t)$$

Operator-valued integro-differential equation, wanna solve it?

To make progress, consider a simpler object: dynamics of density

$$i\hbar\partial_t\hat{n} = i\hbar\partial_t(\hat{\Psi}^\dagger\hat{\Psi}) = \dots?$$



Operator-valued continuity equation

Density dynamics $\rightarrow$ continuity equation (particle is also conserved locally)

$$\partial_t \hat{n}(\mathbf{r}, t) + \nabla \cdot \hat{\mathbf{j}}(\mathbf{r}, t) = 0$$

Particle current operator:

$$\hat{\mathbf{j}}(\mathbf{r}, t) \equiv \frac{\hbar}{2mi} \left[\hat{\Psi}^\dagger(\mathbf{r}, t) \nabla \hat{\Psi}(\mathbf{r}, t) - (\nabla \hat{\Psi}^\dagger(\mathbf{r}, t)) \hat{\Psi}(\mathbf{r}, t) \right]$$

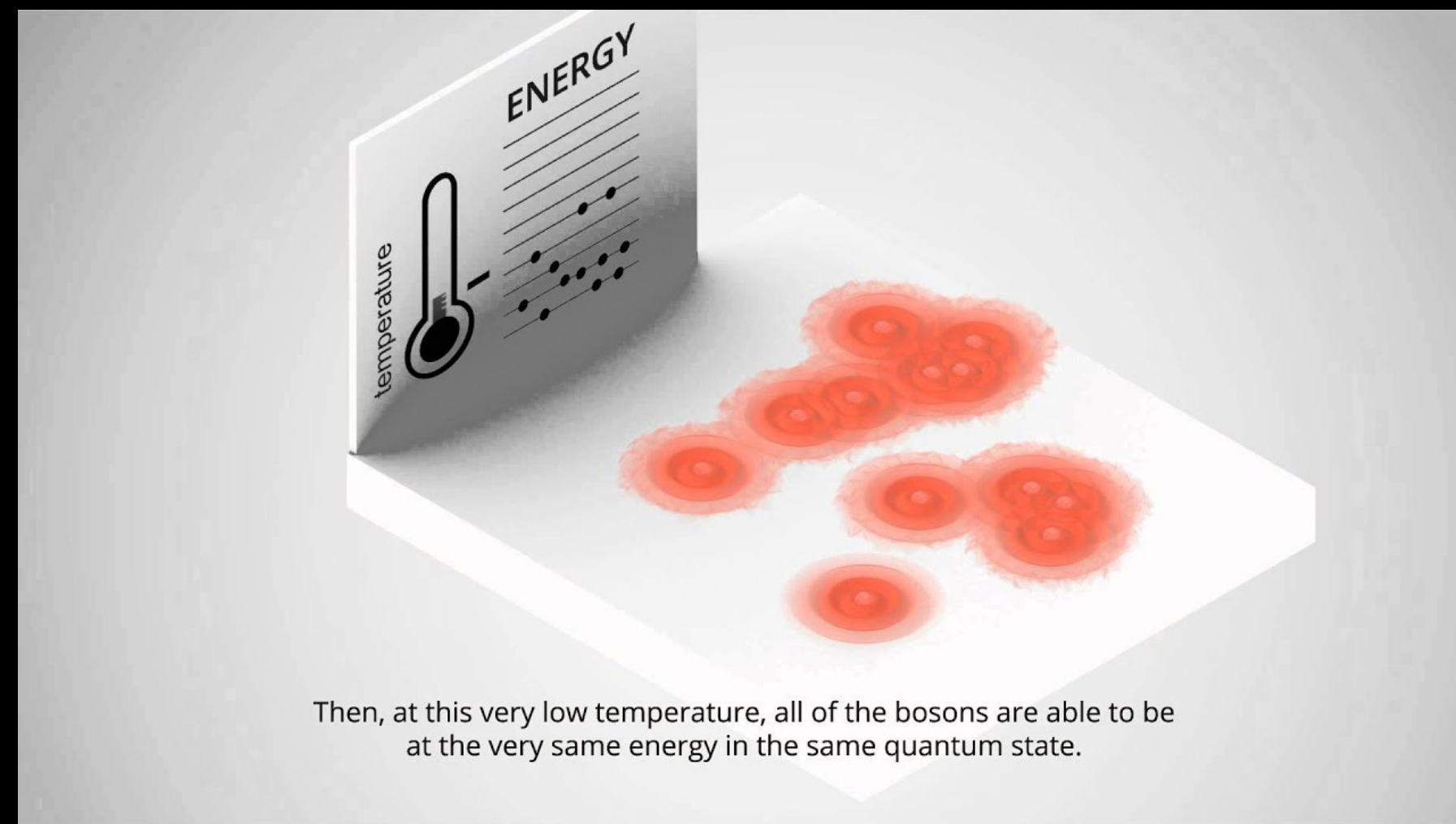
(WARNING ⚠: NOT THE SAME AS CONTINUITY EQUATION FOR SINGLE PARTICLE WAVEFUNCTION)

- **Scalar-valued vs. operator-valued** (governs also the fluctuation around the mean)
- **These are OBSERVABLES:** can actually be measured in experiments, unlike “probability density of a single particle”

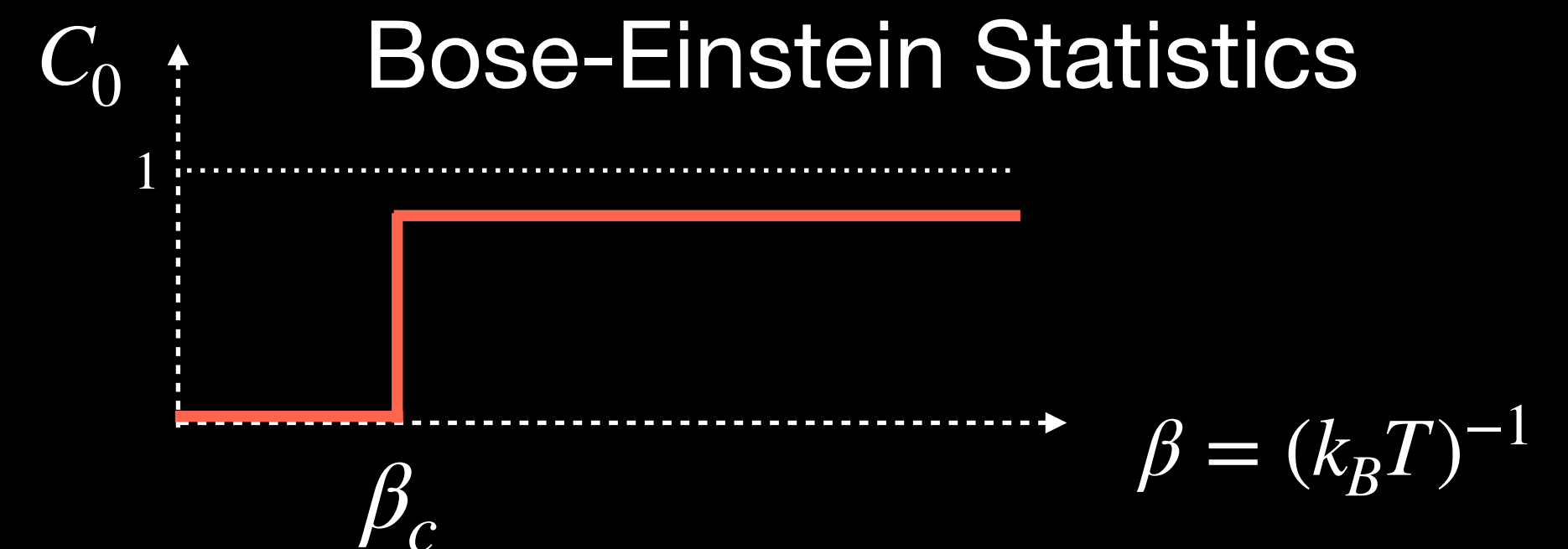
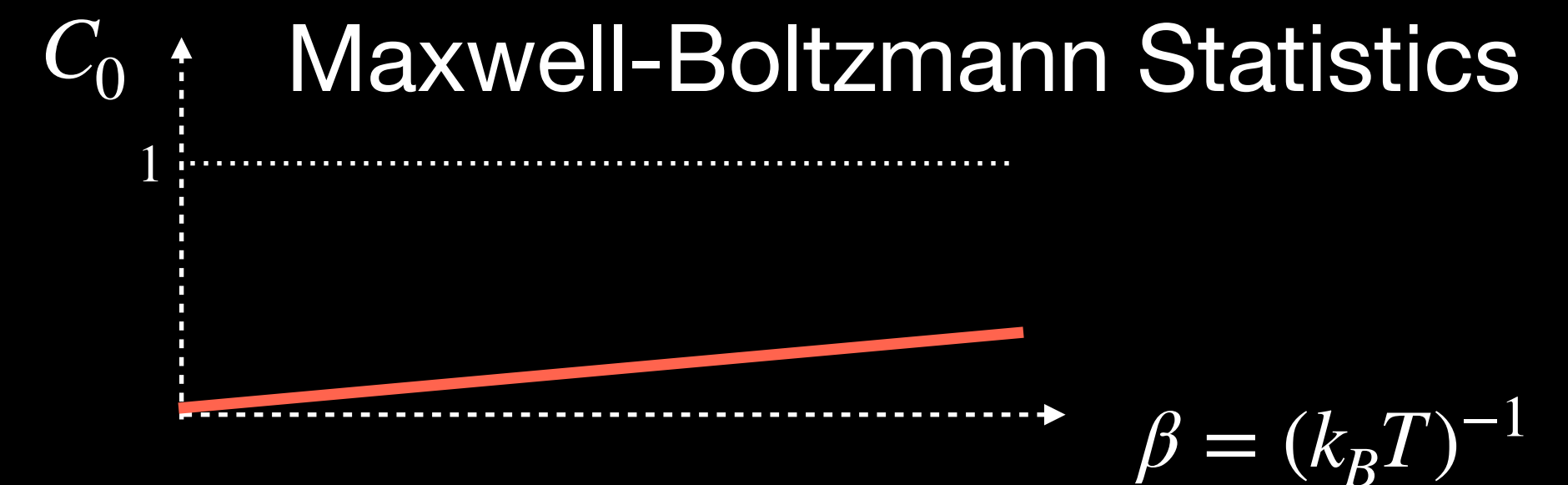
Bose-Einstein Condensate (BEC)

BEC = macroscopic occupation of the single-particle ground state
(some would disagree)

$$C_0 \equiv \lim_{N \rightarrow \infty} \frac{\langle \hat{b}_0^\dagger \hat{b}_0 \rangle}{N} > 0 \leftrightarrow \text{BEC}$$



Intimately connected to
superconductivity and superfluidity



Off-diagonal long-range order (ODLRO)

Second-order correlator

$$G_2(\mathbf{r}, \mathbf{r}') \equiv \langle \hat{\Psi}^\dagger(\mathbf{r}) \hat{\Psi}(\mathbf{r}') \rangle$$

Assume translation-invariant $G_2(|\mathbf{r} - \mathbf{r}'|) \rightarrow$ no external potential $V(\mathbf{r})$

$$\lim_{|\mathbf{r}-\mathbf{r}'|} G_2(|\mathbf{r} - \mathbf{r}'|) \neq 0 \leftrightarrow \text{ODLRO}$$

Ground-state
macroscopic
occupation

Translation invariance +
thermodynamic limit

ODLRO

$$G_2(\mathbf{r}, \mathbf{r}') = \underbrace{N \varphi_0^*(\mathbf{r}) \varphi_0(\mathbf{r}')}_{\propto 1/V} + \sum_{\alpha, \beta} \langle \hat{a}_\alpha^\dagger \hat{a}_\beta \rangle \underbrace{\varphi_\alpha^*(\mathbf{r}) \varphi_\beta(\mathbf{r}')}_{\propto 1/V} \neq 0$$

U(1) Symmetry Breaking

Assume cluster-state decomposition:

$$\lim_{\mathbf{r} \rightarrow \infty} \langle \hat{O}_A(\mathbf{r}) \hat{O}_B(0) \rangle - \langle \hat{O}_A(\mathbf{r}) \rangle \langle \hat{O}_B(0) \rangle = 0.$$

typically satisfied for thermal or ground states of locally interacting Hamiltonian as well as the vacuum states of local quantum field theories

$$G_2(\mathbf{r}) = \langle \hat{\Psi}^\dagger(\mathbf{r}) \rangle \langle \hat{\Psi}^\dagger(0) \rangle \neq 0 \longrightarrow \langle \hat{\Psi}(\mathbf{r}) \rangle = \sqrt{n_0} e^{i\phi} \neq 0$$

U(1) Symmetry: $\hat{\Psi}(\mathbf{r}) \rightarrow e^{i\phi} \hat{\Psi}(\mathbf{r})$ leaves $\hat{H} \rightarrow \hat{H}$

**U(1) Symmetry
Breaking!**

Mean-field theory for BEC

$$C_0 \equiv \lim_{N \rightarrow \infty} \frac{\langle \hat{b}_0^\dagger \hat{b}_0 \rangle}{N} > 0 \leftrightarrow \text{BEC}$$

Translation invariance +
thermodynamic limit

ODLRO

Mean-field approximation

$$\hat{\Psi}(\mathbf{r}) \approx \langle \Psi(\mathbf{r}) \rangle + \delta \hat{\Psi}(\mathbf{r}) \approx \Psi(\mathbf{r})$$

Cluster
decomposition

**U(1) Symmetry
Breaking**

$$i\hbar\partial_t\Psi(\mathbf{r},t) = \left[-\frac{\hbar^2}{2m} + V(\mathbf{r}) \right] \Psi(\mathbf{r},t) + \int d\mathbf{r}' U(\mathbf{r},\mathbf{r}') \Psi^*(\mathbf{r}',t) \Psi(\mathbf{r}',t) \Psi(\mathbf{r},t)$$

Scalar integro-differential equation

Gross-Pitaevskii Equation (GPE)

Dilute atomic gas ~ interaction is well-approximated by contact interaction

$$U(\mathbf{r}, \mathbf{r}') = g\delta(\mathbf{r} - \mathbf{r}')$$

Time-dependent GPE

$$i\hbar\partial_t\Psi(\mathbf{r}, t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r}, t)\right]\Psi(\mathbf{r}, t) + g|\Psi(\mathbf{r}, t)|^2\Psi(\mathbf{r}, t)$$

$$V(\mathbf{r}, t) = V(\mathbf{r})$$

$$\Psi(\mathbf{r}, t) = \Psi(\mathbf{r})e^{-i\mu t/\hbar}$$

Time-independent GPE

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + (V(\mathbf{r}) - \mu)\right]\Psi(\mathbf{r}) + g|\Psi(\mathbf{r})|^2\Psi(\mathbf{r}) = 0$$

Beyond Mean-field Theory



Science Magazine, 22 Dec. 1995

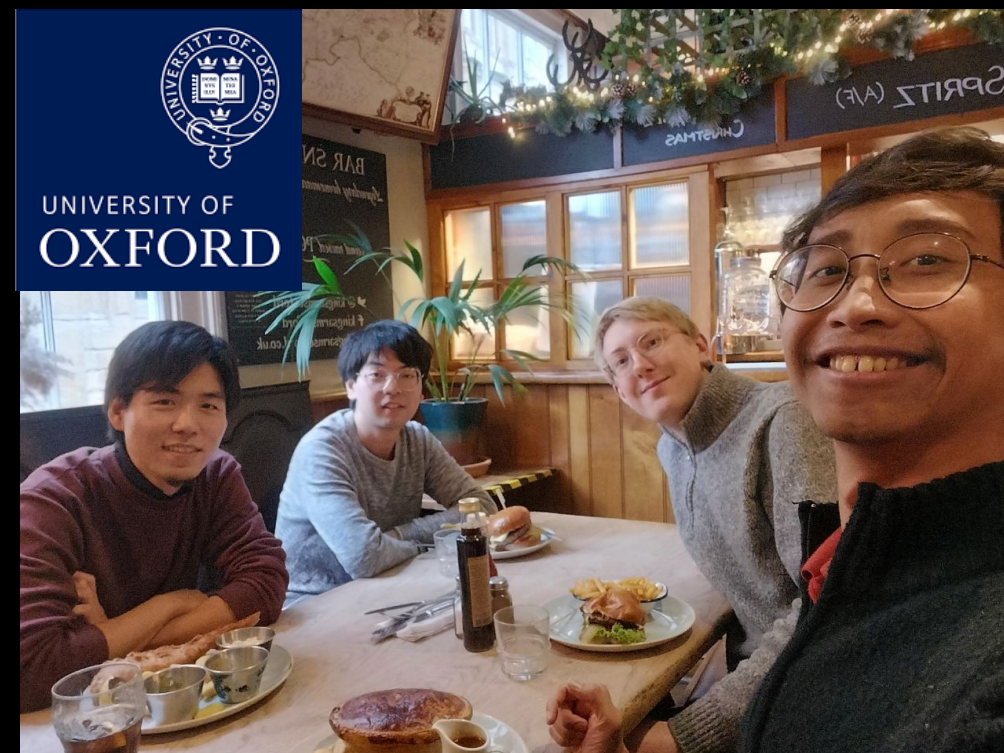
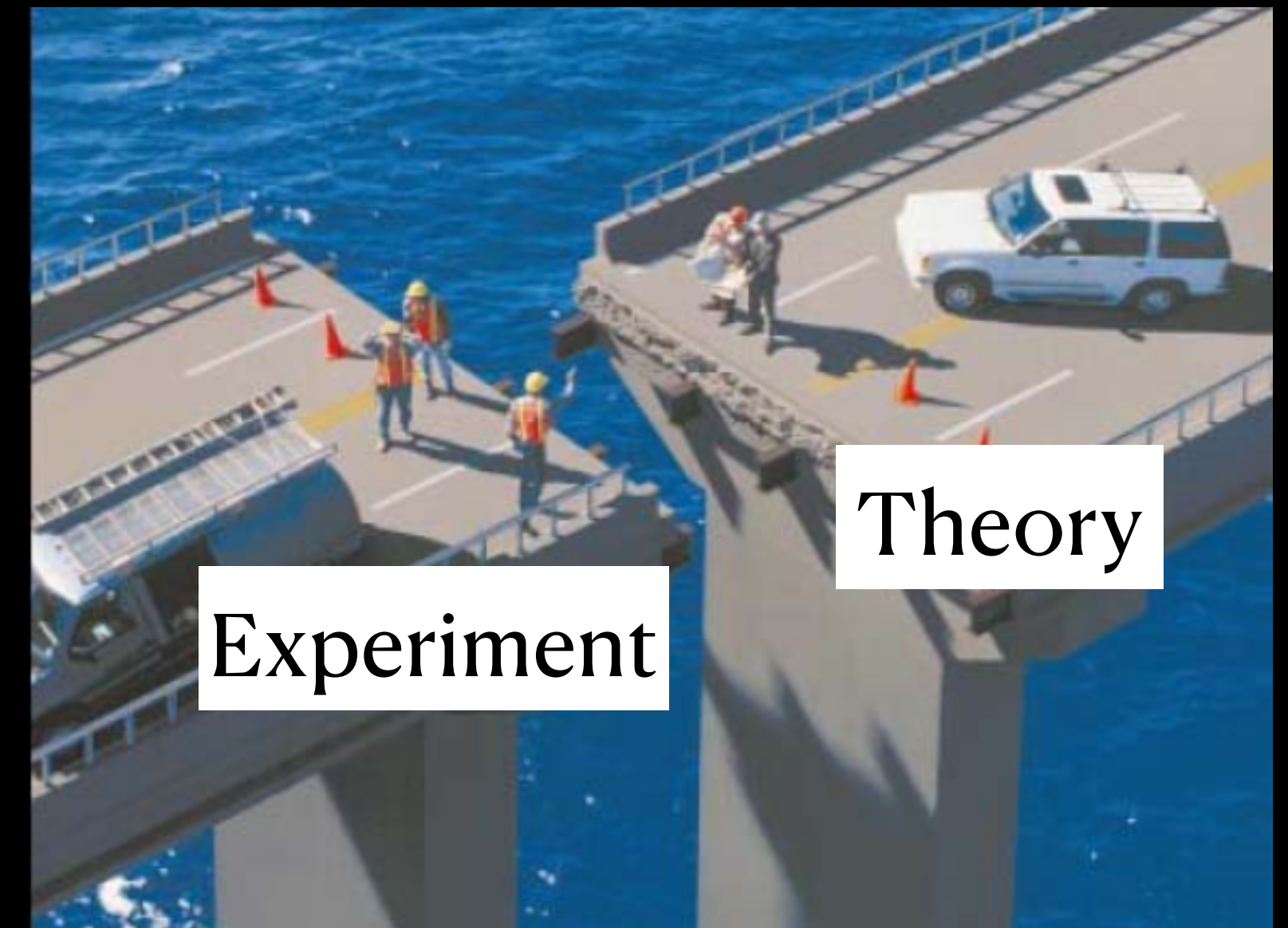
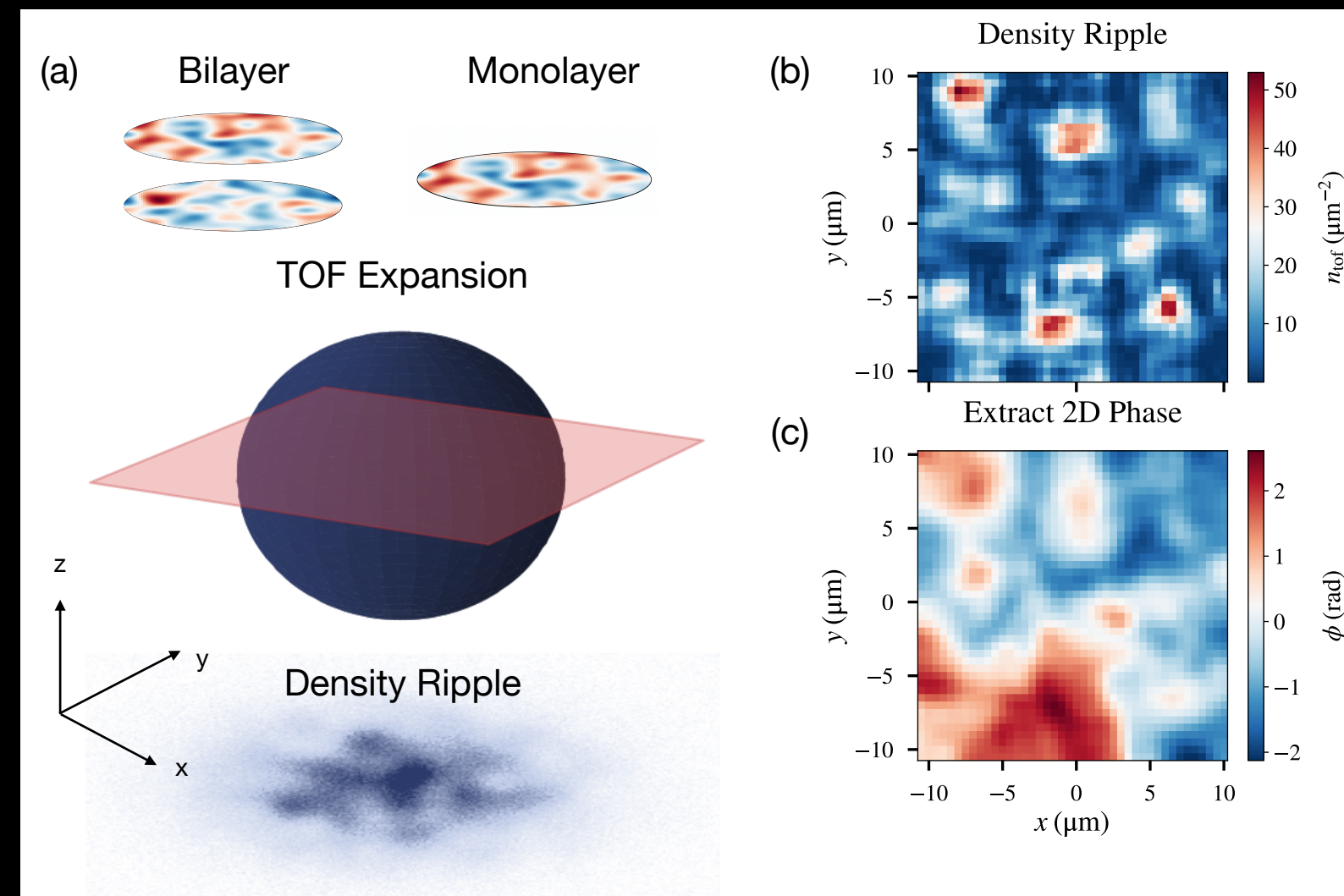
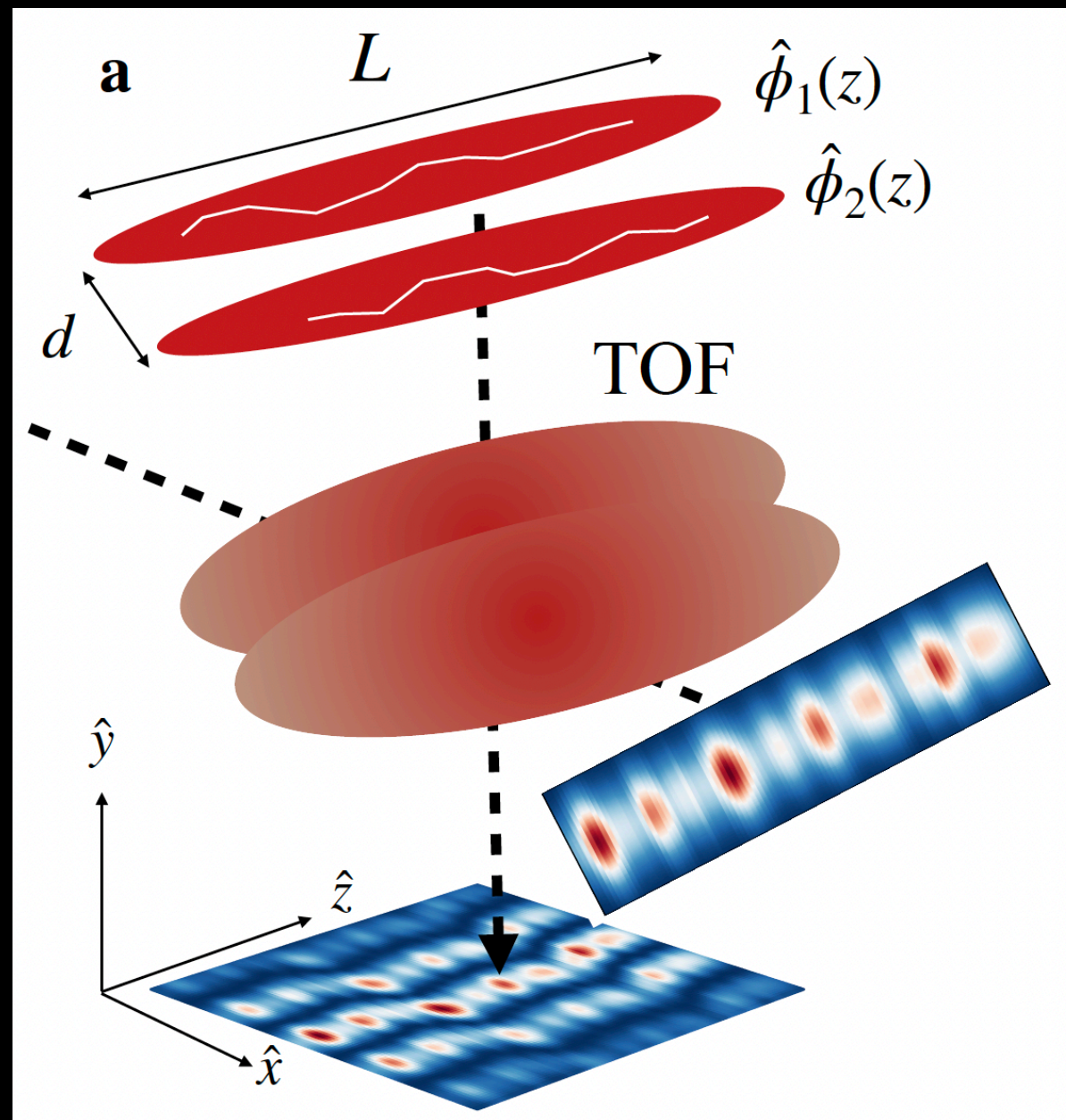
In thermodynamic limit, continuous symmetry breaking is NOT allowed at any temperature in 1D and at finite temperature in 2D (Mermin-Wagner-Hohenberg theorem)

$$\hat{\Psi}(\mathbf{r}) = e^{i\hat{\phi}(\mathbf{r})} \sqrt{n_0(\mathbf{r}) + \delta\hat{n}(\mathbf{r})}$$

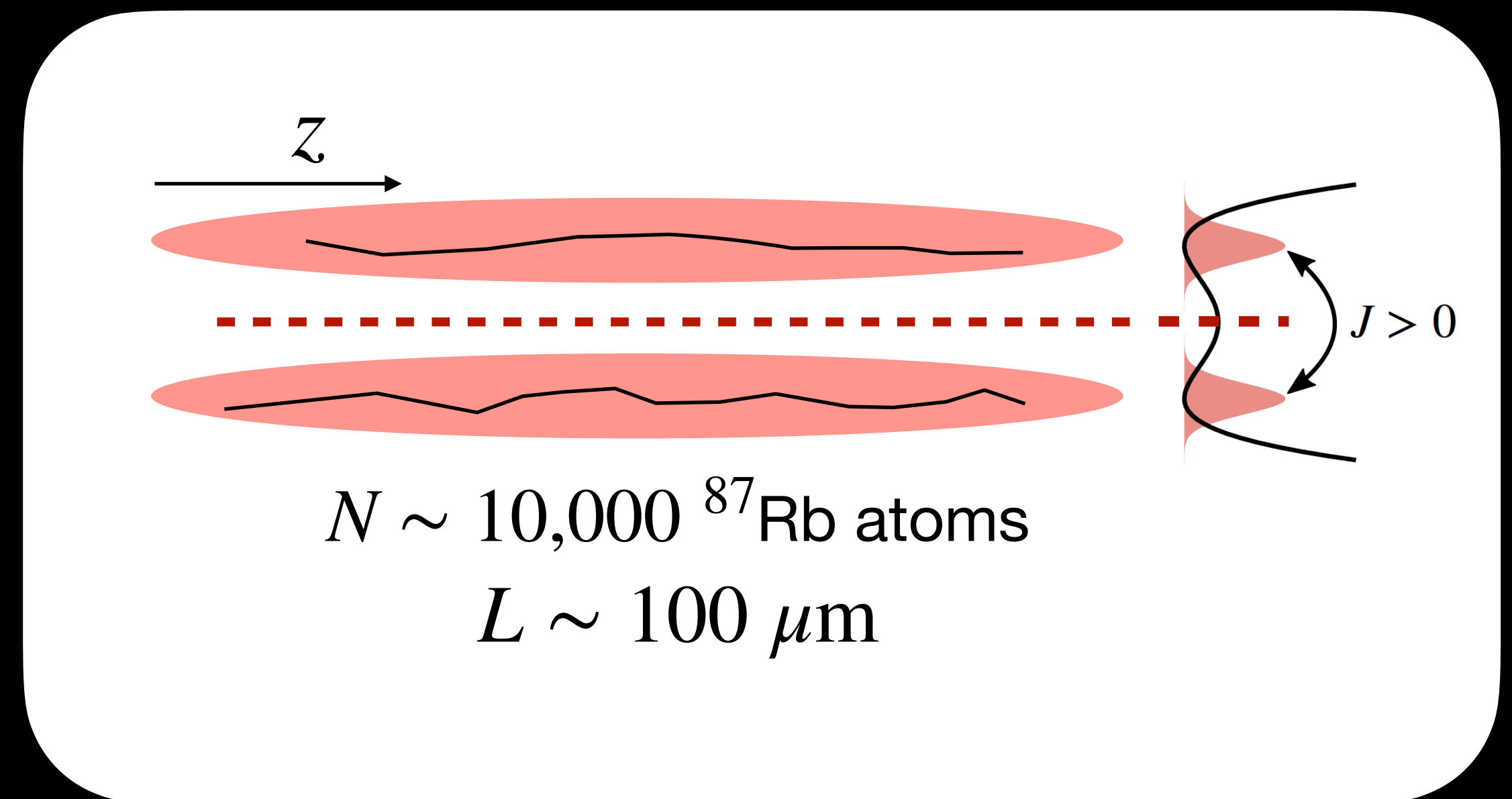
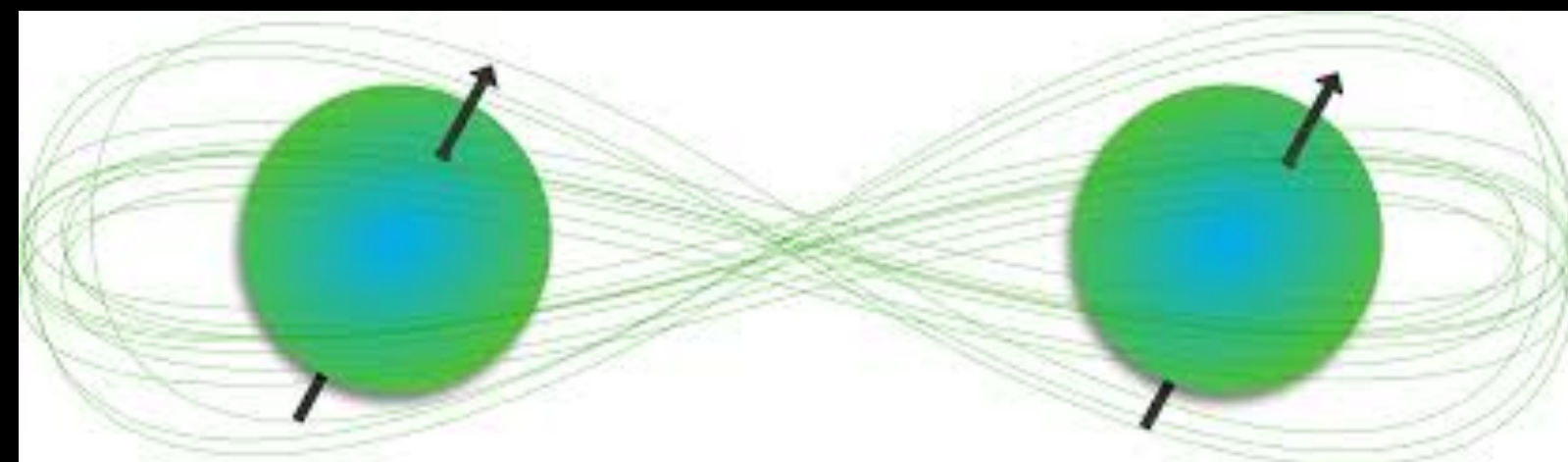
$$[\delta\hat{n}(\mathbf{r}), \hat{\phi}(\mathbf{r}')] = i\delta(\mathbf{r} - \mathbf{r}')$$

In 1D/2D, effects of (quantum/thermal) fluctuation due to interaction is enhanced

My Research: Quantum Field Reconstruction in 1D/2D Bose Gases



Pushing the Limit of Entanglement & Emergence of Classicality



Can we demonstrate entanglement between two “macroscopic” objects?

Pushing the Limit of Entanglement & Emergence of Classicality

arXiv > cond-mat > arXiv:2508.20533

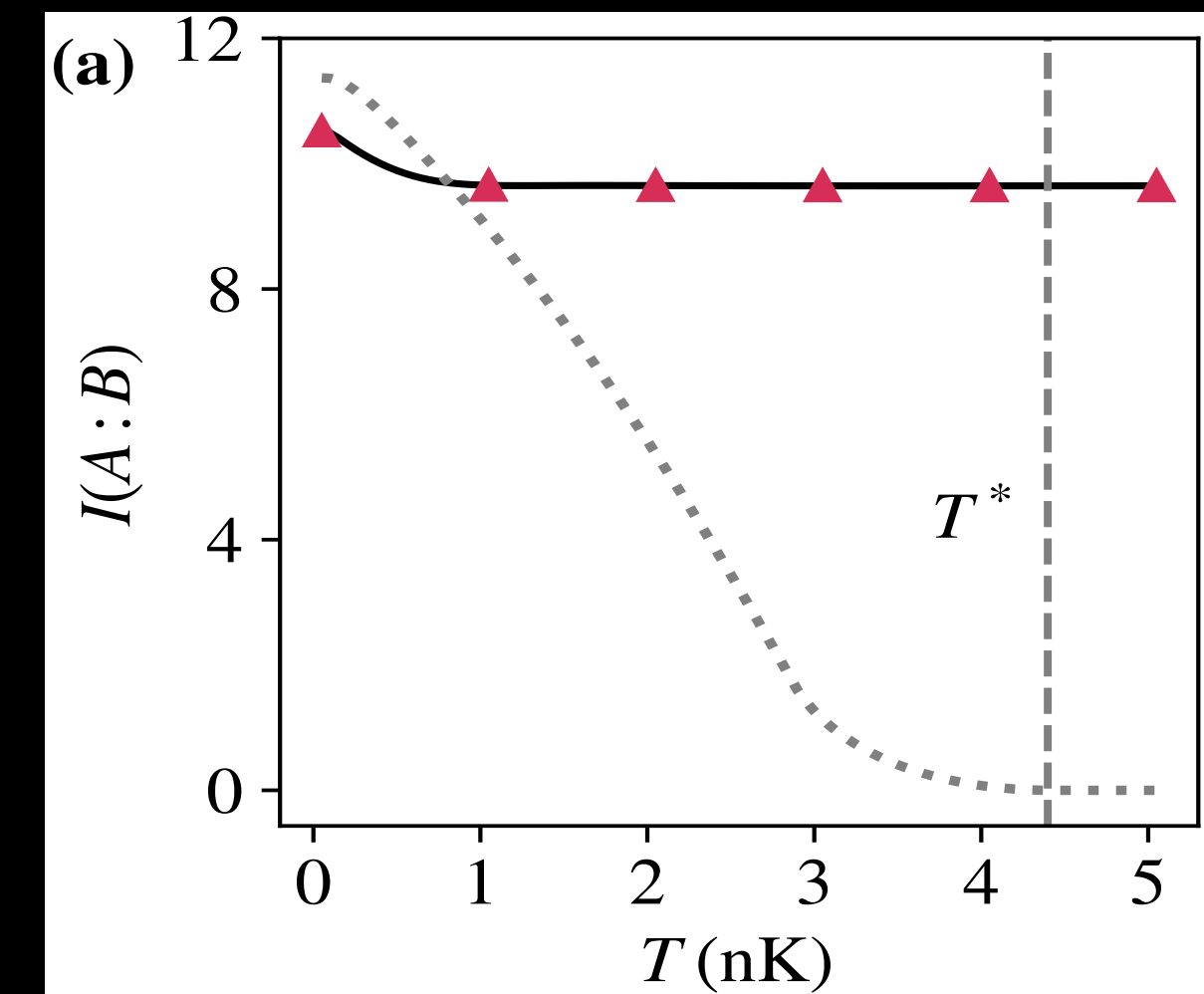
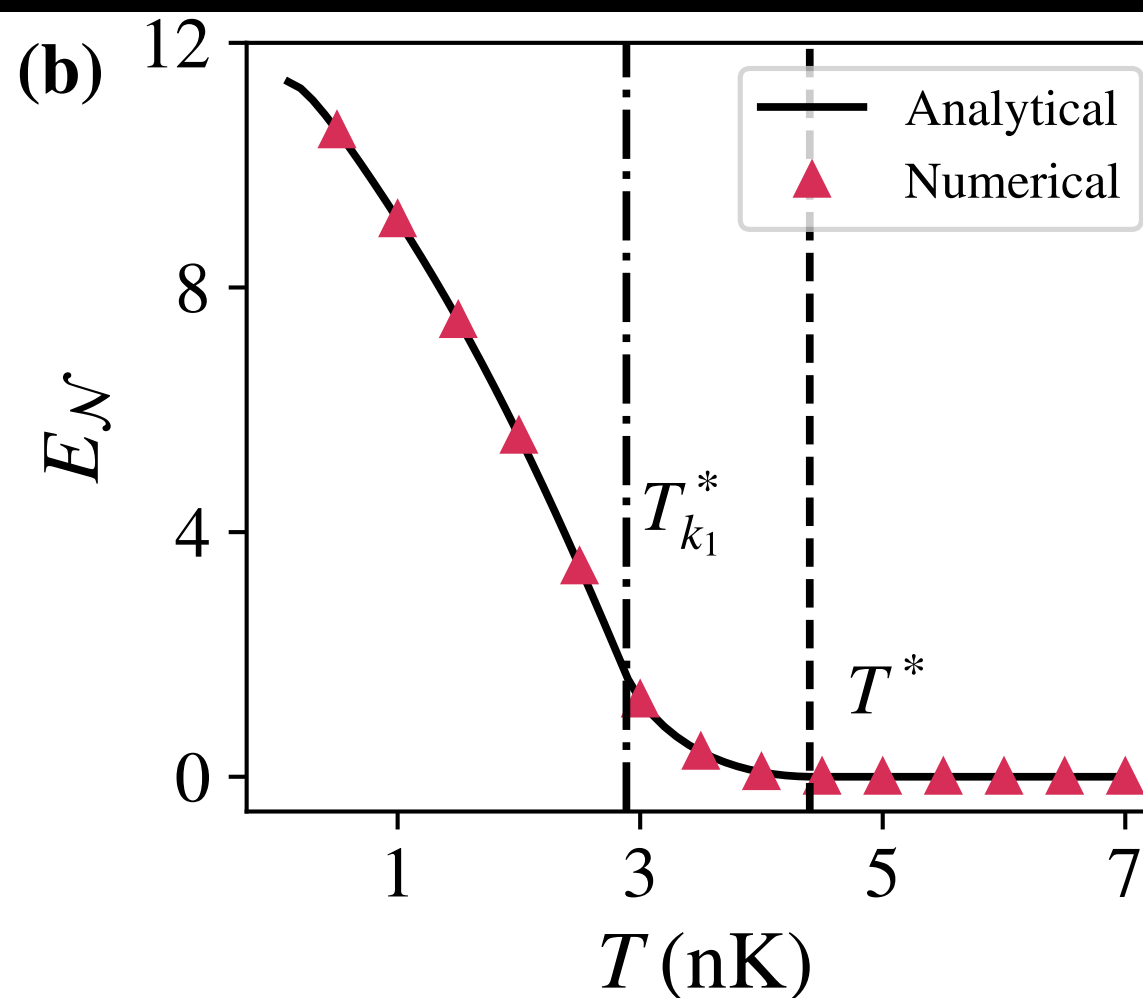
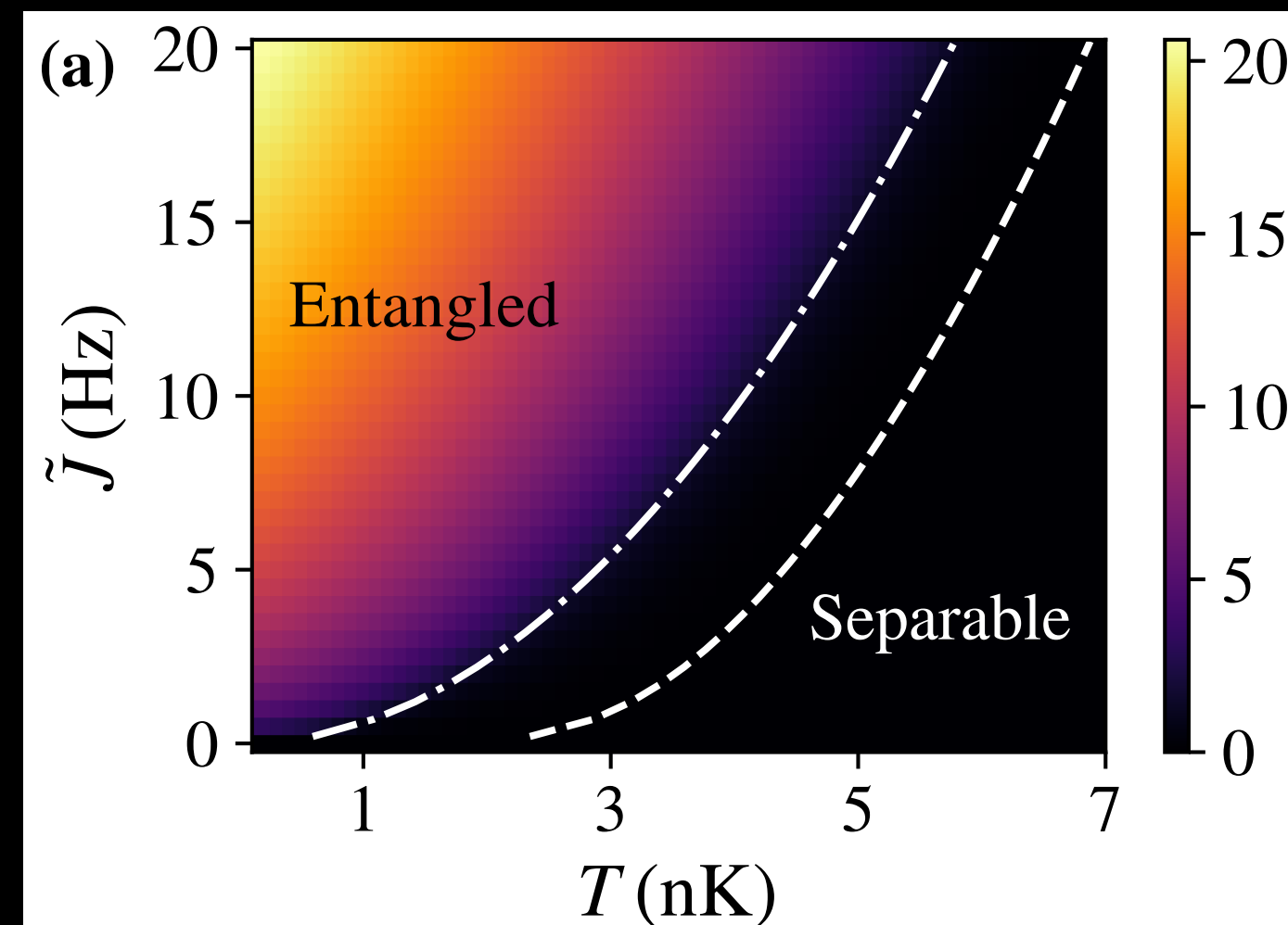
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Extensive entanglement between coupled Tomonaga-Luttinger liquids in and out of equilibrium

Taufiq Murtadho, Marek Gluza, Nelly H. Y. Ng



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